

Program.cs

```
1 using System;
2 using System.Linq;
3 using Week3;
4
5 class Program
6 {
7     static ICoordinatorRepository coordinator = null!;
8
9     static void Main()
10    {
11        var conn = Environment.GetEnvironmentVariable("SIS_CONN") ??
"Server=localhost;Port=3306;Database=sis;Uid=appuser;Pwd=BalkanMadman123IsZURAB;";
12        using var repo = new MySqlCoordinatorRepository(conn);
13        coordinator = repo;
14
15        SeedSampleDataIfEmpty();
16
17        while (true)
18        {
19            Console.WriteLine();
20            Console.WriteLine("Are you a (1) Coordinator or (2) Student? (q to
quit)");
21            Console.Write("Choose: ");
22            var mode = Console.ReadLine();
23            if (mode == "q") break;
24            if (mode == "1") CoordinatorMenu();
25            else if (mode == "2") StudentMenu();
26        }
27    }
28
29    static void SeedSampleDataIfEmpty()
30    {
31        if (coordinator.Organizations.Count > 0) return;
32        SeedSampleData();
33    }
34
35    static void SeedSampleData()
36    {
37        var org = new Organization("Acme Research", "Main 1", "Enschede") { Email =
"info@acme.nl", Url = "https://acme.example" };
38        var cp = new ContactPerson { FirstName = "Jan", LastName = "Jansen", Email
= "j.jansen@acme.nl", JobTitle = "Manager" };
39        org.Contacts.Add(cp);
40        coordinator.AddOrganization(org);
41
42        var internship = new Internship(org.OrganizationId, org.Name, "Smart
Sensors", "Short desc", "Long description", new Period(2025, 1),
InternshipCategory.Research);
```

```
43     internship.Contacts.Add(cp);
44     coordinator.AddInternship(internship);
45 }
46
47 static void CoordinatorMenu()
48 {
49     while (true)
50     {
51         Console.Clear(); // clear before showing menu
52         Console.WriteLine("Coordinator Menu:");
53         Console.WriteLine("(1) List Orgs");
54         Console.WriteLine("(2) Add Org");
55         Console.WriteLine("(3) List Internships");
56         Console.WriteLine("(4) Withdraw Internship");
57         Console.WriteLine("(5) Mark Completed");
58         Console.WriteLine("(6) Remove Student");
59         Console.WriteLine("(7) Add Internship");
60         Console.WriteLine("(8) Add Student");
61         Console.WriteLine("(9) List Students");
62         Console.WriteLine("(10) List Contacts");
63         Console.WriteLine("(11) Add Contact");
64         Console.WriteLine("(b) Back");
65         Console.Write("Choose: ");
66         var cmd = Console.ReadLine();
67         if (cmd == "b") break;
68
69         if (cmd == "1")
70         {
71             Console.Clear();
72             Console.WriteLine("Organizations:");
73             foreach (var o in coordinator.GetOrganizationsOverview())
74                 Console.WriteLine($"{o.OrganizationId} - {o}");
75             Console.WriteLine();
76             Console.WriteLine("Press Enter to continue...");
77             Console.ReadLine();
78         }
79         else if (cmd == "2")
80         {
81             Console.Clear();
82             Console.Write("Name: "); var name = Console.ReadLine() ?? "";
83             Console.Write("Address: "); var address = Console.ReadLine() ?? "";
84             Console.Write("City: "); var city = Console.ReadLine() ?? "";
85             var org = new Organization(name, address, city);
86             if (!coordinator.AddOrganization(org)) Console.WriteLine("Duplicate
organization.");
87             else
88             {
89                 Console.WriteLine($"Added {org.OrganizationId}");
90                 Console.Write("Add a contact now? (y/n): ");
```

```
91         var add = Console.ReadLine();
92         if (add?.ToLower() == "y")
93         {
94             var cp = PromptContactFields();
95             if (cp != null)
96                 coordinator.AddContactPerson(org.OrganizationId, cp);
97             Console.WriteLine("Contact added.");
98         }
99         Console.WriteLine("Press Enter to continue...");
100        Console.ReadLine();
101    }
102    else if (cmd == "3")
103    {
104        Console.Clear();
105        Console.Write("Year: "); var y = int.Parse(Console.ReadLine() ??
106        "0");
107        Console.Write("Semester (1 or 2): "); var s =
108        int.Parse(Console.ReadLine() ?? "0");
109        var list = coordinator.GetInternshipsOverview(new Period(y, s));
110        Console.Clear();
111        Console.WriteLine("Internships:");
112        foreach (var i in list) Console.WriteLine($"{i.Id} - {i}");
113        Console.WriteLine();
114        Console.WriteLine("Press Enter to continue...");
115        Console.ReadLine();
116    }
117    else if (cmd == "4")
118    {
119        Console.Clear();
120        Console.Write("Internship Id to withdraw: "); var id =
121        Console.ReadLine();
122        if (!string.IsNullOrEmpty(id) &&
123        coordinator.WithdrawInternship(id)) Console.WriteLine("Withdrawn.");
124        else Console.WriteLine("Not found or failed.");
125        Console.WriteLine("Press Enter to continue...");
126        Console.ReadLine();
127    }
128    else if (cmd == "5")
129    {
130        Console.Clear();
131        Console.Write("Internship Id to mark completed: "); var id =
132        Console.ReadLine();
133        Console.Write("Final grade (numeric): "); var raw =
134        Console.ReadLine();
135        if (double.TryParse(raw, out var grade) && !
136        string.IsNullOrEmpty(id) && coordinator.MarkInternshipCompleted(id, grade))
137            Console.WriteLine("Marked completed.");
138        else Console.WriteLine("Failed to mark completed (invalid id or
139        grade).");
```

```
131         Console.WriteLine("Press Enter to continue...");
132         Console.ReadLine();
133     }
134     else if (cmd == "6")
135     {
136         Console.Clear();
137         Console.Write("Internship Id: "); var iid = Console.ReadLine();
138         Console.Write("Student Id to remove: "); var sid =
139         Console.ReadLine();
140         if (!string.IsNullOrEmpty(iid) && !
141         string.IsNullOrEmpty(sid) && coordinator.RemoveStudentFromInternship(sid,
142         iid)) Console.WriteLine("Student removed from internship.");
143         else Console.WriteLine("Failed to remove student (check ids).");
144         Console.WriteLine("Press Enter to continue...");
145         Console.ReadLine();
146     }
147     else if (cmd == "7")
148     {
149         Console.Clear();
150         AddInternshipPrompt();
151         Console.WriteLine("Press Enter to continue...");
152         Console.ReadLine();
153     }
154     else if (cmd == "8")
155     {
156         Console.Clear();
157         AddStudentPrompt();
158         Console.WriteLine("Press Enter to continue...");
159         Console.ReadLine();
160     }
161     else if (cmd == "9")
162     {
163         Console.Clear();
164         var students = coordinator.Students.ToList();
165         if (!students.Any()) { Console.WriteLine("No students
166         registered."); }
167         else
168         {
169             Console.WriteLine("Students:");
170             foreach (var s in students) Console.WriteLine($"{s.StudentId} -
171             {s.FirstName} {s.LastName}");
172         }
173         Console.WriteLine();
174         Console.WriteLine("Press Enter to continue...");
175         Console.ReadLine();
176     }
177     else if (cmd == "10")
178     {
179         Console.Clear();
```

```
175         Console.WriteLine("List contacts by: (1) All organizations (2)  
Specific organization");  
176         var choice = Console.ReadLine();  
177         if (choice == "1")  
178         {  
179             foreach (var org in coordinator.GetOrganizationsOverview())  
180             {  
181                 Console.WriteLine();  
182                 Console.WriteLine($"Org: {org.Name}  
({org.OrganizationId})");  
183                 var cps = coordinator.GetContactPersonsByO-  
rganization(org.OrganizationId);  
184                 foreach (var cp in cps) PrintContact(cp);  
185             }  
186         }  
187         else  
188         {  
189             var orgs = coordinator.GetOrganizationsOverview().ToList();  
190             for (int i = 0; i < orgs.Count; i++) Console.WriteLine($"{i+1}.  
{orgs[i].Name} ({orgs[i].OrganizationId})");  
191             Console.Write("Select organization number: ");  
192             if (int.TryParse(Console.ReadLine(), out var idx) && idx >= 1  
&& idx <= orgs.Count)  
193             {  
194                 var org = orgs[idx - 1];  
195                 var cps = coordinator.GetContactPersonsByO-  
rganization(org.OrganizationId);  
196                 Console.Clear();  
197                 Console.WriteLine($"Contacts for {org.Name}:");  
198                 foreach (var cp in cps) PrintContact(cp);  
199             }  
200         }  
201         Console.WriteLine();  
202         Console.WriteLine("Press Enter to continue...");  
203         Console.ReadLine();  
204     }  
205     else if (cmd == "11")  
206     {  
207         Console.Clear();  
208         var orgs2 = coordinator.GetOrganizationsOverview().ToList();  
209         if (!orgs2.Any()) { Console.WriteLine("No organizations. Add org  
first."); Console.WriteLine("Press Enter to continue..."); Console.ReadLine();  
continue; }  
210         for (int i = 0; i < orgs2.Count; i++) Console.WriteLine($"{i+1}.  
{orgs2[i].Name} ({orgs2[i].OrganizationId})");  
211         Console.Write("Select organization number to add contact to: ");  
212         if (!int.TryParse(Console.ReadLine(), out var sel) || sel < 1 ||  
sel > orgs2.Count) { Console.WriteLine("Invalid selection.");  
Console.WriteLine("Press Enter to continue..."); Console.ReadLine(); continue; }  
213         var chosenOrg = orgs2[sel - 1];
```

```
214         var contact = PromptContactFields();
215         if (contact != null &&
coordinator.AddContactPerson(chosenOrg.OrganizationId, contact))
Console.WriteLine("Contact added.");
216         else Console.WriteLine("Failed to add contact.");
217         Console.WriteLine("Press Enter to continue...");
218         Console.ReadLine();
219     }
220     else
221     {
222         Console.Clear();
223         Console.WriteLine("Unknown command. Please choose a menu number or
'b' to go back.");
224         Console.WriteLine("Press Enter to continue...");
225         Console.ReadLine();
226     }
227 }
228 }
229
230 static ContactPerson? PromptContactFields()
231 {
232     Console.Write("First name: "); var fn = Console.ReadLine();
233     Console.Write("Last name: "); var ln = Console.ReadLine();
234     Console.Write("Email: "); var email = Console.ReadLine();
235     Console.Write("Job title: "); var job = Console.ReadLine();
236     Console.Write("Phone: "); var phone = Console.ReadLine();
237     Console.Write("Department: "); var dept = Console.ReadLine();
238     var cp = new ContactPerson
239     {
240         FirstName = string.IsNullOrWhiteSpace(fn) ? null : fn,
241         LastName = string.IsNullOrWhiteSpace(ln) ? null : ln,
242         Email = string.IsNullOrWhiteSpace(email) ? null : email,
243         JobTitle = string.IsNullOrWhiteSpace(job) ? null : job,
244         PhoneNumber = string.IsNullOrWhiteSpace(phone) ? null : phone,
245         Department = string.IsNullOrWhiteSpace(dept) ? null : dept
246     };
247     return cp;
248 }
249
250 static void PrintContact(ContactPerson cp)
251 {
252     Console.WriteLine($" {cp.Id} - {cp.FirstName ?? ""} {cp.LastName ?? ""}");
253     Console.WriteLine($"      Email: {cp.Email ?? "-"} Job: {cp.JobTitle ??
254     "-"} Phone: {cp.PhoneNumber ?? "-"} Dept: {cp.Department ?? "-}");
255 }
256
257 static void AddInternshipPrompt()
258 {
259     Console.WriteLine();
260     var orgs = coordinator.GetOrganizationsOverview().ToList();
```

```
260     if (!orgs.Any()) { Console.WriteLine("No organizations found. Add an  
organization first."); return; }  
261  
262     Console.WriteLine("Select organization:");  
263     for (int i = 0; i < orgs.Count; i++) Console.WriteLine($"{i+1}.  
{orgs[i].Name} ({orgs[i].OrganizationId})");  
264     var sel = Console.ReadLine();  
265     if (!int.TryParse(sel, out var idx) || idx < 1 || idx > orgs.Count) {  
Console.WriteLine("Invalid selection."); return; }  
266     var org = orgs[idx - 1];  
267  
268     Console.Write("Title: "); var title = Console.ReadLine() ?? "";  
269     Console.Write("Short description: "); var shortDesc = Console.ReadLine() ??  
"";  
270     Console.Write("Long description: "); var longDesc = Console.ReadLine() ??  
"";  
271     Console.Write("Year: "); var year = int.TryParse(Console.ReadLine(), out  
var y) ? y : DateTime.Now.Year;  
272     Console.Write("Semester (1 or 2): "); var sem =  
int.TryParse(Console.ReadLine(), out var s) ? s : 1;  
273  
274     Console.WriteLine("Category: 0=Research, 1=Minor, 2=Engineering (enter  
number)");  
275     var catInput = Console.ReadLine();  
276     InternshipCategory category = InternshipCategory.Research;  
277     if (int.TryParse(catInput, out var cval) &&  
Enum.IsDefined(typeof(InternshipCategory), cval))  
278         category = (InternshipCategory)cval;  
279  
280     var internship = new Internship(org.OrganizationId, org.Name, title,  
shortDesc, longDesc, new Period(year, sem), category);  
281  
282     Console.Write("Add existing contact(s) from organization to this  
internship? (y/n): ");  
283     if (Console.ReadLine()?.ToLower() == "y")  
284     {  
285         var contacts = coordinator.GetContactPersonsByO-  
rganization(org.OrganizationId).ToList();  
286         for (int i = 0; i < contacts.Count; i++) Console.WriteLine($"{i+1}.  
{contacts[i].FirstName} {contacts[i].LastName} ({contacts[i].Id})");  
287         Console.WriteLine("Enter comma-separated numbers to add (or blank):");  
288         var choices = Console.ReadLine();  
289         if (!string.IsNullOrEmpty(choices))  
290         {  
291             foreach (var part in choices.Split(',',  
StringSplitOptions.RemoveEmptyEntries))  
292             {  
293                 if (int.TryParse(part.Trim(), out var n) && n >= 1 && n <=  
contacts.Count)  
294                     internship.Contacts.Add(contacts[n - 1]);  
295             }  
}
```

```
296         }
297     }
298
299     coordinator.AddInternship(internship);
300     Console.WriteLine();
301     Console.WriteLine($"Added internship {internship.Id}");
302 }
303
304 static void AddStudentPrompt()
305 {
306     Console.WriteLine();
307     Console.Write("Student Id: "); var sid = Console.ReadLine();
308     if (string.IsNullOrEmpty(sid)) { Console.WriteLine("Id required.");
return; }
309     Console.Write("First name: "); var fn = Console.ReadLine();
310     Console.Write("Last name: "); var ln = Console.ReadLine();
311     var student = new Student(sid, fn ?? "", ln ?? "");
312     if (coordinator.AddStudent(student)) Console.WriteLine($"Student
{student.StudentId} added/updated.");
313     else Console.WriteLine("Failed to add student.");
314 }
315
316 static void StudentMenu()
317 {
318     while (true)
319     {
320         Console.Clear(); // clear before showing student menu
321         Console.WriteLine("Student Menu:");
322         Console.WriteLine("(1) Browse internships");
323         Console.WriteLine("(2) Register");
324         Console.WriteLine("(b) Back");
325         Console.Write("Choose: ");
326         var cmd = Console.ReadLine();
327         if (cmd == "b") break;
328         if (cmd == "1")
329         {
330             Console.Clear();
331             Console.Write("Year: "); var y = int.Parse(Console.ReadLine() ??
"0");
332             Console.Write("Semester: "); var s = int.Parse(Console.ReadLine()
?? "0");
333             var list = coordinator.GetInternshipsOverview(new Period(y,
s)).ToList();
334             Console.Clear();
335             var idx = 1;
336             foreach (var i in list) Console.WriteLine($"{idx++}.
{i.OrganizationName} | {i.Title} | {i.ShortDescription} (Id: {i.Id})");
337             Console.WriteLine();
338             Console.WriteLine("Enter number to view details or blank to
return:");
```



```
339         var sel = Console.ReadLine();
340         if (int.TryParse(sel, out int n) && n > 0 && n <= list.Count)
341         {
342             var chosen = list[n - 1];
343             Console.Clear();
344             Console.WriteLine($"Title: {chosen.Title}\nOrganization:
{chosen.OrganizationName}\nPeriod: {chosen.Period}\nCategory: {chosen.Category}");
345             Console.WriteLine();
346             Console.WriteLine("Contacts for this internship:");
347             var contacts = coordinator.GetContactPersonsFor-
Internship(chosen.Id).ToList();
348             if (!contacts.Any()) Console.WriteLine(" (no contacts)");
349             else foreach (var c in contacts) PrintContact(c);
350
351             Console.WriteLine();
352             Console.Write("Enter your student id to apply (blank to
cancel): "); var sid = Console.ReadLine();
353             if (string.IsNullOrEmpty(sid)) {
Console.WriteLine("Cancelled."); Console.WriteLine("Press Enter to continue...");
Console.ReadLine(); continue; }
354             Console.Write("First name: "); var fn = Console.ReadLine();
355             Console.Write("Last name: "); var ln = Console.ReadLine();
356             var student = new Student(sid, fn ?? "", ln ?? "");
357             if (coordinator.AssignStudentToInternship(student, chosen.Id))
358             {
359                 Console.WriteLine("Assigned (provisional). Contact info
above for follow-up.");
360                 if (contacts.Any())
361                 {
362                     Console.WriteLine();
363                     Console.WriteLine("Contact(s) to reach out to:");
364                     foreach (var c in contacts) PrintContact(c);
365                 }
366             }
367             else Console.WriteLine("Cannot assign (possibly conflict or
duplicate).");
368
369             Console.WriteLine();
370             Console.WriteLine("Press Enter to continue...");
371             Console.ReadLine();
372         }
373     }
374     else if (cmd == "2")
375     {
376         Console.Clear();
377         AddStudentPrompt();
378         Console.WriteLine("Press Enter to continue...");
379         Console.ReadLine();
380     }
381     else
```

```
382         {
383             Console.Clear();
384             Console.WriteLine("Unknown command. Please choose a menu number or
'b' to go back.");
385             Console.WriteLine("Press Enter to continue...");
386             Console.ReadLine();
387         }
388     }
389 }
390 }
391
392
393
```